AIMS
Airbus Material Specification

Glass Fibre Reinforced Epoxy Prepreg

1581, 7581 and 7781 style Glass Fabric
(8-Harness Satin, 300 g/m^2 dry fibre area mass)

180°C Cure Temperature

Material Specification

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Contents

1 Scope
2 Application
3 Normative References
4 Technical Requirements
5 Designation
6 Technical Specification
7 Incoming Inspection

1 Scope

This Material Specification defines the requirements of epoxy resin reinforced with 300g/m^2 8-harness satin glass fibre fabric (1581/7581/7781 style) with a nominal cure temperature of 180°C for aerospace application. The requirements specified in this Material Specification shall not be used as stress allowables.

Unless otherwise specified by the drawing, the order or the inspection schedule, the present specification shall be used in conjunction with the referenced documents.

2 Application

This material is used for the manufacture of structural and non-structural applications.
Materials to be delivered according to the relevant ABS.

3 Normative References

This Airbus specification incorporates by dated or undated reference provisions from other publications. All normative references cited at the appropriate places in the text are listed hereafter. For dated references, subsequent amendments to or revisions of any of these publications apply to this Airbus specification only when incorporated in it by amendment of revision. For undated references, the latest issue of the publication referred to shall be applied.

AIMS 05-02-000Part 1	Airbus Material Specification - Glass fibre reinforced thermosetting prepreg Technical Specification - General Requirements
AIMS 05-02-000 Part 3	Airbus Material Specification - Glass fibre reinforced thermosetting prepreg Fabric / Epoxy resin - 180°C curing class - Technical Specification - Qualification and batch release testing
AIMS 05-02-000Part 100	Airbus Material Specification - Glass fibre reinforced thermosetting prepreg Technical Specification – Acceptance testing & storage life prolongation
ABS 5009	Airbus Standard – EP-Prepreg. Glass fibre fabric pre-impregnated with epoxy resin
AITM 1-0001	Airbus Test Method – Fibre reinforced plastics – Determination of mechanical degradation due to chemical paint strippers
AITM 1-0002	Airbus Test Method – Fibre reinforced plastics – Determination of in plane shear properties (± 45° tensile tests)
AITM 1-0003	Airbus Test Method – Fibre reinforced plastics – Determination of the glass transition temperatures
AITM 1-0010	Airbus Test Method – Fibre reinforced plastics – Determination of compression strength after impact
AITM 1-0019	Airbus Test Method – Fibre reinforced plastics – Determination of tensile lap shear strength of composite joints
AITM 2-0037	Airbus Test Method – Fluid tightness test for sandwich specimens and structures
AITM 3-0001	Airbus Test Method – Fibre Reinforced plastics – Analysis of thermoset systems by high performance liquid chromatography
AITM 3-0002	Airbus Test Method – Analysis of non metallic materials (uncured) by differential scanning calorimetry
AITM 3-0003	Airbus Test Method – Analysis of non-organic compounds by infrared spectroscopy
AITM 3-0004	Airbus Test Method – Determination of gel time and viscosity
AITM 3-0006	Airbus Test Method – Fibre reinforced plastics – Determination of material degradation due to chemical products
AITM 3-0008	Airbus Test Method – Determination of the extent of cure by differential scanning calorimetry
EN 2155-12	Aerospace Series – Test Methods for transparent materials for aircraft glazing – Part 12 Determination of linear thermal expansion

EN 2329	Aerospace Series – Textile glass fibre preimpregnates – Test method for the determination of mass per unit area
EN 2330	Aerospace Series – Textile glass fibre preimpregnates – Test method for the determination of volatile content
EN 2331	Aerospace Series - Textile glass fibre preimpregnates - Test method for the determination of the resin and fibre content of woven textile glass and mass of fibre per unit area
EN 2332	Aerospace Series - Textile glass fibre preimpregnates - Test methods for the determination of the resin flow
EN 2377	Aerospace Series - Glass fibre reinforced plastics - Determination of apparent interlaminar shear strength
EN 2743	Aerospace Series - Fibre reinforced plastics - Standards procedures for conditioning prior to testing unaged materials
EN 2747	Aerospace Series - Glass fibre reinforced plastics - Tensile test
EN 2823	Aerospace Series - Fibre reinforced plastics - Determination of the effect of exposure to humid atmosphere on physical and mechanical characteristics
EN 2850	Aerospace Series - Carbon thermosetting resin unidirectional laminates - Compression test parallel to the fibre direction
EN 3615	Aerospace Series - Fibre reinforced plastics - Procedure for the determination of the conditions of exposure to humid atmosphere and the determination of moisture absorption.
ISO 1183 Method A	Plastics - Method for determining the density and relative density of non-cellular plastics
ISO 1889	Reinforcement Yarns - Determination of the linear density
AMS-C-9084	Cloth - Glass finished for resin laminates

4 Technical Requirements

4.1 General

Specification values are minimum except where stated.

4.1.1 Material Description

Table 1: Material Description

No.	Characteristic		Requirements
1	Material Type		Glass fibre fabric reinforced epoxy prepreg
2	Formulation	Fibre	Glass fibre, Type E
		Fabric	Style 1581/7581/7781, Satin Weave AMS-C-9084
		Matrix	Modified epoxy resin
		Sizing/Finish	Epoxy compatible
3	Resin Content by weight, uncured		As defined in table 4
4	Material Code		As defined in table 4

4.1.2 Material Application

The different possible applications of glass fibre fabric epoxy prepregs are defined in the technical specification AIMS 05-02-000 Part 3.
This specification defines the requirements for two types of material application: Structural and Non-structural

Note: All glass fibre prepregs qualified for a structural application can be used for non-structural application without any additional tests.

4.1.3 Work Life and storage life

Worklife and storage life requirements are defined in table 2.
Specific requirements for storage conditions are listed in the relevant IPS.

Table 2: Worklife and storage conditions

No.	Properties	Shop Condition	Symbol	Unit	Requirements
1	Storage life at -18°C	-	-	months	≥ 12
2	Work Life (Tack life) 1)	30°C / 75% R.H.	-	days	≥ 10
		(25±2)°C / (45±10)% R.H	-	days	≥10
Notes: 1) The prepreg can be exposed to an additional minimum of 5 days at the above mentioned shop condition provided that the lay-up is fully bagged and ready for use.					

4.1.4 Processing Data

The curing cycle shall be agreed between the Manufacturer and Airbus at the onset of the material qualification program and shall be used to produce all test specimens for qualification testing. The laminates for test purposes shall be cured in an autoclave or press. The processing conditions are shown in Table 3.

The agreed curing cycle shall be specified in the relevant IPS and shall be used for the manufacture of all the batch release test specimens.

Table 3: Processing conditions

No.	Cure Parameter	Unit	Requirements
Cure Type: Autoclave			
1	Heating rate	°C/min	0.2 – 5.0
2	Cure Temperature	°C	180 +10/-5
3	Cure Time	Minute	120 - 240
4	Cure pressure	MPa	0.3 – 1.1
5	Cooling rate	°C/min	0.2 – 3.5
6	Vacuum	MPa	0.005 - 0.1
Note: The use of an intermediate temperature and low pressure dwell is permissable if agreed between the Manufacturer and the Purchaser.			

4.2 Specific

4.2.1 Physical-Chemical Properties

Uncured and cured properties, see table 4.

4.2.2 Mechanical Properties

Cured laminate properties, see table 5

Table 4: Physical-Chemical properties for all applications

Test No.	Property	Test Method	Unit	Requirements [1]		
Material Code				**AIMS 05-02-003**		
				A	**B**	**C**
Application				**NS**		**ST**
Uncured						
1	Prepreg areal weight	EN2329	g/m^2	470 ± 35	500 ± 35	470 ± 35
2	Fibre areal weight [2]	EN2331	g/m^2	300 ± 20	300 ± 20	300 ± 20
3	Resin density	ISO 1183A	g/cm^3	See relevant IPS		
4	Fibre density	ISO 1889	g/cm^3	2.54 ± 0.05		
5	Volatile content [2]	EN2330	%	≤ 2		
6	Resin content [2]	EN2331	%	37 ± 3	40 ± 3	37 ± 3
7	Physio-Chemical Tests: HPLC	AITM 3-0001A	-	(3)		
	Physio-Chemical Tests: DSC	AITM 3-0002 10°C/min				
	Physio-Chemical Tests: IR	AITM 3-0003, solvent: defined during qualification				
	Physio-Chemical Tests: Gel Time	AITM 3-0004 B or C				
	Physio-Chemical Tests: Viscosity	AITM 3-0004A 2°C/min				
8	Resin Flow	EN2332	%	Value to be reported		
9	Tack	As defined in AIMS 05-02-000 part 1	-	(2)		
Cured						
10	CTE [4] (warp and weft)	EN2155-12	°C^{-1}	Value to be reported		
11	Water uptake [5]	EN3615 (70°C, 85% RH)	%	≤ 2.0		
12	Extent of cure	AITM 3-0008	%	≥ 90		

Notes:

The stated results are average values
Application: NS = Non-Structural; ST = Structural (see AIMS05-02-000 Part3)

1) 3 specimens per batch
2) Method to be agreed before qualification between purchaser and supplier
3) See AIMS 05-02-000 part 1 § 5.4.3, and recorded in the IPS
4) Coefficient of Thermal Expansion
5) Additional water uptake conditions can be agreed during qualification process if necessary

Table 5: Mechanical properties for structural application

Test No.	Property		Test Method	Unit	Conditioning[1)]	Test Temp. (°C)	Requirements [3)] X	Requirements [3)] X_{min}
Material Code							**AIMS 05-02-003A, B, C**	
1	Glass transition temperature, dry and wet [3)]	Tg onset	AITM 1-0003	°C	dry	N/A	140	140
					wet		110	110
		Tg loss			dry		Value to be reported	Value to be reported
					wet			
Material Code							**AIMS 05-02-003C**	
2a	Tensile strength 0° warp		EN2747 (Type 3) [2)]	MPa	dry	-55	500	450
						RT	410	370
						70	330	300
					wet	70	315	295
						90 [4)]	280	250
2b	Tensile Modulus 0° warp			GPa	dry	-55	27 ± 3	
						RT	24 ± 3	
						70	24 ± 3	
					wet	70	24 ± 3	
						90 [4)]	24 ± 3	
2c	Poissons ratio 0/90°				dry	RT	Value to be reported	Value to be reported
						70		
						90 [4)]		
3a	Tensile strength 90° weft			MPa	dry	-55	470	440
						RT	395	375
					wet	70	270	250
						90 [4)]	260	235
3b	Tensile modulus 90° weft			GPa	dry	-55	25 ± 3	
						RT	24 ± 3	
					wet	70	23 ± 3	
						90 [4)]	22 ± 3	
4a	Compression strength 0° warp		EN 2850B	MPa	dry	-55	800	725
						RT	660	600
						70	550	500
						90 [4)]	470	430
					wet	70	470	425
						90 [4)]	400	360
4b	Compression modulus 0° warp			GPa	dry	-55	26 ± 4	
						RT	25 ± 3	
						70	25 ± 3	
						90 [4)]	25 ± 3	
					wet	70	25 ± 3	
						90 [4)]	25 ± 3	
5a	Compression strength 90° weft			MPa	dry	RT	490	445
					wet	70	320	310
						90 [4)]	320	305

(Continued)

Table 5: Mechanical properties for structural applications (Continued)

Test No.	Property	Test Method	Unit	Conditioning[1)]	Test Temp. (°C)	Requirements [3)] X	Requirements [3)] X_{min}
Material Code						**AIMS 05-02-003C**	
5b	Compression modulus 90° weft	EN2850B	GPa	dry	RT	24±3	
				wet	70	24±3	
					90 [4)]	23±3	
6a	In plane shear strength 45°	AITM 1-0002	MPa	dry	-55	110	100
					RT	94	85
					70	88	80
					90 [4)]	86	78
					120 [4)]	77	70
				wet	RT	83	75
					70	77	70
					90 [4)]	66	60
					120 [4)]	53	48
6b	In plane shear modulus 45°	AITM 1-0002	GPa	dry	-55	5.7 ± 1.0	
					RT	4.8 ± 1.0	
					70	3.9 ± 1.0	
					90 [4)]	3.7 ± 1.0	
					120 [4)]	3.2 ± 1.0	
				wet	RT	4.3 ± 1.0	
					70	3.5 ± 1.0	
					90 [4)]	3.1 ± 1.0	
					120 [4)]	1.7 ± 1.0	
7	ILSS 0° warp	EN2377	MPa	dry	-55	88	80
					RT	68	62
					70	59	54
					90 [4)]	55	50
					120 [4)]	46.2	42
				wet	RT	49.5	45
					70	39.6	36
					90 [4)]	36.3	33
					120 [4)]	27.5	25
8	Tensile lap shear 0° warp [6)]	AITM 1-0019 S1	MPa	dry	-55	Value to be reported	Value to be reported
					RT		
					70		
				wet	RT		
					70		
					90 [4)]		
9	CAI	AITM 1-0010	MPa	dry	RT	Value to be reported	Value to be reported
10	Fluid tightness [4)]	AITM 2-0037	-	-	-	Value to be reported	Value to be reported

continued

Table 5: Mechanical properties for structural applications (Concluded)

Test No.	Property	Test Method	Unit	Conditioning[1)]	Test Temp. (°C)	Requirements [3)]	
						X	X_{min}
Material Code						**AIMS 05-02-003C**	
Resistance to aggressive fluids with ILSS warp [5)6) 7)]							
11a	Reference	EN2377	MPa	dry	RT	68	62
				wet	70	59	54
11b	Hydraulic fluid (70°C/1000h)		MPa	dry	70	48	44
11c	Hydraulic fluid plus water mix (50°C / 1000h)		MPa	dry	70	43	40
11d	Fuel (RT / 1000h)		MPa	dry	70	52	47
11e	MEK (RT / 1h)		MPa	dry	RT	58	53
12	Paint stripper (AITM 1-0001, paint stripper: Turco 1270-5)		MPa	dry	RT	Value to be reported	Value to be reported

Notes:

Requirements for properties 2 to 5 are calculated with the nominal ply thickness of 0.250mm related to a fibre volume content of 46%. Other requirements are calculated with a real thickness.

1) Dry conditions according to EN2743, Wet conditions according to EN2823
2) Type 2 specimens (untabbed) may be used in agreement with the purchaser
3) 6 specimens per batch unless otherwise specified with the Purchaser. X = minimum average, Xmin = minimum individual.
4) Test not mandatory
5) AIMS 05-02-000 part 1 § 4.4.6, AITM 3-0006 and AITM 1-0001 (only required if no data available from other qualification programmes using same resin/fibre/size combination)
6) Tensile lap shear strength test will have to be tested only in case of unacceptable failure mode with ILSS test

7) Test laminate to be manufactured with peel ply (as defined in the relevant QTP)

5 Designation

See relevant ABS.

6 Technical Specification

The general technical requirements and responsibilities corresponding to this Material Specification are described in the Technical Specification AIMS05-02-000.

7 Incoming Inspection

Prior to use, the materials qualified against this AIMS shall undergo “Technical Incoming Inspection” in accordance with AIMS05-02-000 Part 100’

RECORD OF REVISIONS

Issue	Clause modified	Description of modification
1 03/06		New standard
2 03/04	Entire doc	Adaption to new 'Glass Prepreg' architecture tree (family 05-02-000) Update of tests requirements
3 12/06	Entire doc	New NS and ST codes added. Update of material allowables. Update of document format.
4 12/13	Chapter 3 Chapter 4.1.1 Chapter 7	Part 100 TS added & AMS-C-9084 standard added Fabric formulation requirements standard modified Chapter 7 defined to link to Part 100 specification for Incoming Inspection